

# The Quartic Tower: Exact Radicals

and the Minimal Quartics of the Rungs  $N = 15/30$

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The “Dynamic Principle” program — the hodge-laboratory repository

Standalone theorem monograph · Monograph 19/20

## 1 Setting

The levels  $N = 15$  and  $N = 30$  are the constructible rungs of the ladder:  $15 = 3 \cdot 5$  is a product of distinct Fermat primes, and the regular pentadecagon is constructible by straightedge and compass (Gauss–Wantzel). Accordingly  $2 \cos(2\pi/15)$  has degree 4 over  $\mathbb{Q}$  ( $\varphi(15)/2 = 4$ ), and  $b_{\text{Ch}}(15)$ ,  $b_{\text{Ch}}(30)$  are expressible in square radicals — by a two-storey tower  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{5}) \subset \mathbb{Q}(\sqrt{5}, \beta)$ . This monograph fixes the complete exact layer of v1.2: the closed radicals, the minimal quartics, the Galois structure of the tower, and the machine certificate — the integer identity check in  $\mathbb{Z}[\sqrt{5}][\beta]$ .

**Lemma 1** (doubling permutes the classes). *The set  $C = \{1, 2, 4, 7\}$  contains one representative of each pair of classes  $\{\pm k\}$  of invertible residues modulo 15. Multiplication by 2 permutes  $C$  (up to changing the representative):  $1 \mapsto 2 \mapsto 4 \mapsto 8 \equiv -7 \mapsto 7 \mapsto 14 \equiv -1 \mapsto 1$ .*

*Proof.* Direct check:  $2 \cdot 1 = 2$ ,  $2 \cdot 2 = 4$ ,  $2 \cdot 4 = 8 \equiv 8$ , and  $8 \equiv -7 \pmod{15}$ ; further  $2 \cdot 7 = 14 \equiv -1$ . The classes  $\{\pm 1\}$ ,  $\{\pm 2\}$ ,  $\{\pm 4\}$ ,  $\{\pm 7\}$  are cycled into one another — this is precisely the cyclic group  $\text{Gal}(\mathbb{Q}(\zeta_{15})^+/\mathbb{Q}) \cong (\mathbb{Z}/15)^\times / \{\pm 1\} \cong C_4$  generated by doubling.  $\square$

**Lemma 2** (Ramanujan sums). *For  $x_k = \zeta^k + \zeta^{-k}$ ,  $\zeta = e^{2\pi i/15}$ ,  $k \in C$ :  $s_1 = \sum x_k = 1$  and  $s_2 = \sum_{i < j} x_i x_j = -4$ .*

*Proof.*  $s_1 = \sum_{k \in C} (\zeta^k + \zeta^{-k}) = \sum_{j \in U} \zeta^j$ , where  $U = (\mathbb{Z}/15)^\times$  runs over all eight units; this is the Ramanujan sum  $c_{15}(1) = \mu(15) = \mu(3)\mu(5) = 1$ . Next,  $x_k^2 = 2 + 2 \cos(4\pi k/15) = 2 + x_{2k}$ , so by Lemma 1  $\sum_{k \in C} x_k^2 = 8 + s_1 = 9$ , and  $s_2 = \frac{s_1^2 - \sum x_k^2}{2} = \frac{1-9}{2} = -4$ .  $\square$

**Theorem 3** (the tower and the radicals). *Put  $\beta = \sqrt{30 - 6\sqrt{5}}$  and  $\beta' = \sqrt{30 + 6\sqrt{5}}$  (positive roots). Then*

$$2 \cos \frac{2\pi}{15} = \frac{1 + \sqrt{5} + \beta}{4}, \quad 2 \cos \frac{2\pi}{30} = \frac{\sqrt{5} - 1 + \beta'}{4},$$

*and  $\beta\beta' = 12\sqrt{5}$  is an exact identity, so that  $\beta' = 12\sqrt{5}/\beta$  and both radicals live in one quartic field  $K = \mathbb{Q}(\sqrt{5}, \beta) = \mathbb{Q}(\sqrt{5}, \beta') = \mathbb{Q}(\zeta_{15})^+ = \mathbb{Q}(\zeta_{30})^+$ . The field  $K$  is totally real and cyclic:  $\text{Gal}(K/\mathbb{Q}) \cong C_4$ , and  $\mathbb{Q}(\sqrt{5})$  is its unique quadratic subfield.*

*Proof. The forms.* For  $x = (1 + \sqrt{5} + \beta)/4$  we have  $(4x - 1 - \sqrt{5})^2 = \beta^2 = 30 - 6\sqrt{5}$ ; expanding,  $16x^2 - 8(1 + \sqrt{5})x + (6 + 2\sqrt{5}) = 30 - 6\sqrt{5}$ , i.e.  $x^2 = \varphi x + \gamma$  with  $\varphi = \frac{1+\sqrt{5}}{2}$  and  $\gamma = \frac{3-\sqrt{5}}{2}$ :  $x$  is quadratic over  $\mathbb{Q}(\sqrt{5})$ . Moreover  $x \notin \mathbb{Q}(\sqrt{5})$ : otherwise the quadratic  $z^2 - \varphi z - \gamma$  would have a root in  $\mathbb{Q}(\sqrt{5})$ , and its discriminant  $\Delta_z = \varphi^2 + 4\gamma = \frac{3(5-\sqrt{5})}{2}$  would have norm  $N(\Delta_z) = \frac{9}{4} \cdot 20 = 45$ , a rational square ( $N(w^2) = N(w)^2 \in \mathbb{Q}^2$ ) — but 45 is not a square. Hence  $[\mathbb{Q}(\sqrt{5}, x) : \mathbb{Q}(\sqrt{5})] = 2$  and  $[\mathbb{Q}(\sqrt{5}, x) : \mathbb{Q}] = 4$ . Since  $x = 2 \cos(2\pi/15)$  lies in  $\mathbb{Q}(\zeta_{15})^+$  and  $\varphi(15)/2 = 4$ , the fields coincide:  $K = \mathbb{Q}(\zeta_{15})^+$ . For the second form:  $\zeta_{30} = -\zeta_{15}^8$  (an order check), so  $\mathbb{Q}(\zeta_{30}) = \mathbb{Q}(\zeta_{15})$ , and the analogous expansion  $(4x' - \sqrt{5} + 1)^2 = \beta'^2$  gives  $x' = (\sqrt{5} - 1 + \beta')/4 = 2 \cos(\pi/15) = 2 \cos(2\pi/30)$ .

*The tower pairing.*  $\beta^2 \beta'^2 = (30 - 6\sqrt{5})(30 + 6\sqrt{5}) = 900 - 180 = 720 = (12\sqrt{5})^2$ , and  $\beta, \beta' > 0$ ; therefore  $\beta\beta' = 12\sqrt{5}$  exactly, whence  $\beta' = 12\sqrt{5}/\beta \in K$ .

*Cyclicity.*  $\text{Gal}(\mathbb{Q}(\zeta_{15})^+/\mathbb{Q}) \cong (\mathbb{Z}/15)^\times/\{\pm 1\}$ ; the element 2 has order 4 modulo 15, so the group is cyclic of order 4. The automorphism  $\sigma: \zeta \mapsto \zeta^2$  sends  $\sqrt{5} = \zeta_5 + \zeta_5^4 - \zeta_5^2 - \zeta_5^3$  (where  $\zeta_5 = \zeta^3$ ) to  $-\sqrt{5}$ ; hence  $\sigma(\beta)^2 = \sigma(\beta^2) = 30 + 6\sqrt{5} = \beta'^2$  and, by the pairing,  $\sigma(\beta) = \pm\beta'$  — the sign is pinned:  $\sigma$  swaps  $\beta \leftrightarrow \beta'$  up to sign, while  $\langle \sigma^2 \rangle$  fixes  $\mathbb{Q}(\sqrt{5})$  — the unique quadratic subfield (the group  $C_4$  has a unique subgroup of order 2). The discriminant of the field: by the conductor–discriminant formula  $\text{disc } K = f_4 \cdot f_4' \cdot f_2 = 15 \cdot 15 \cdot 5 = 1125 = 3^2 \cdot 5^3$  — the very number that equals the volume  $\text{vol}_h = 1125$  of monograph 16/20.  $\square$

**Theorem 4** (the minimal quartics). *The minimal polynomials are*

$$P_{15}(x) = x^4 - x^3 - 4x^2 + 4x + 1 \text{ for } 2 \cos \frac{2\pi}{15}, \quad P_{30}(x) = x^4 + x^3 - 4x^2 - 4x + 1 \text{ for } 2 \cos \frac{2\pi}{30},$$

and  $P_{30}(x) = P_{15}(-x)$  — the level-30 quartic is the reflection of the level-15 quartic. The coefficients of  $P_{15}$  assemble from Ramanujan sums and the tower norm: the constant term  $s_4 = \prod x_k = 1$  and the third coefficient  $s_3 = -4$  (Newton).

*Proof.* The quartic with roots  $x_1, \dots, x_4$  (the classes  $C$ ) reads  $x^4 - s_1x^3 + s_2x^2 - s_3x + s_4$ . By Lemma 2 already  $s_1 = 1$ ,  $s_2 = -4$ . *The tower norm:* by Theorem 3  $K \supset \mathbb{Q}(\sqrt{5})$  and the norm splits:  $x_1x_3 = (\zeta + \zeta^{-1})(\zeta^4 + \zeta^{-4}) = (\zeta^5 + \zeta^{-5}) + (\zeta^3 + \zeta^{-3}) = -1 + 2 \cos \frac{2\pi}{5} = -1 + \frac{\sqrt{5}-1}{2} = \frac{\sqrt{5}-3}{2}$ ; then  $s_4 = N_{K/\mathbb{Q}}(x_1) = N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}\left(\frac{\sqrt{5}-3}{2}\right) = \frac{\sqrt{5}-3}{2} \cdot \frac{-\sqrt{5}-3}{2} = 1$ . *Newton:*  $p_3 = \sum x_k^3 = \sum_{k \in C} [\zeta^{3k} + \zeta^{-3k}] + 3x_k$ ; here  $3k \bmod 15$  runs over  $\{3, 6, 12, 6\}$  — pairs of fifth parts:  $\sum_{k \in C} (\zeta^{3k} + \zeta^{-3k}) = 2 \cdot 2 \cos \frac{2\pi}{5} + 2 \cdot 2 \cos \frac{4\pi}{5} = (\sqrt{5} - 1) - (1 + \sqrt{5}) = -2$ , so  $p_3 = -2 + 3s_1 = 1$ , and Newton's identity gives  $s_3 = \frac{p_3 - s_1p_2 + s_2p_1}{3} = \frac{1 - 9 - 4}{3} = -4$  ( $p_2 = \sum x_k^2 = 9$ ,  $p_1 = s_1 = 1$ ). Altogether  $P_{15}(x) = x^4 - x^3 - 4x^2 + 4x + 1$ ; it is the minimal polynomial because  $[K : \mathbb{Q}] = 4$  is already established.

*The reflection.* The roots of  $P_{30}$  are  $2 \cos(\pi k/15)$  for  $k \in \{1, 7, 11, 13\}$  (the classes  $\{\pm k\}$  modulo 30). For  $m \in C$ :  $-2 \cos \frac{2\pi m}{15} = 2 \cos(\pi - \frac{2\pi m}{15}) = 2 \cos \frac{\pi(15-2m)}{15}$ , and  $15 - 2m$  runs over  $\{13, 11, 7, 1\}$  — exactly the classes of  $P_{30}$ . Hence the root set of  $P_{30}$  is  $\{-x : x \text{ a root of } P_{15}\}$ , i.e.  $P_{30}(x) = P_{15}(-x) = x^4 + x^3 - 4x^2 - 4x + 1$ .

*Corollary (the  $b_{\text{Ch}}$  forms).*  $b_{\text{Ch}}(15) = 1 - \frac{x}{2} = \frac{7-\sqrt{5}-\beta}{8}$ ,  $b_{\text{Ch}}(30) = 1 - \frac{x'}{2} = \frac{9-\sqrt{5}-\beta'}{8}$ .  $\square$

**Theorem 5** (the exact layer: an integer certificate). (a) *The element  $x = m/4$  with  $m = 1 + \sqrt{5} + \beta$  satisfies  $P_{15}$  identically: the expansion of  $256 P_{15}(m/4)$  in the tower  $\mathbb{Z}[\sqrt{5}][\beta]$  ( $\beta^2 = 30 - 6\sqrt{5}$  — an integer element!) collapses to the zero quadruple  $(0, 0, 0, 0)$  — a pure integer identity without a single floating-point operation; likewise  $256 P_{30}((\sqrt{5} - 1 + \beta')/4) = 0$  in  $\mathbb{Z}[\sqrt{5}][\beta']$ .* (b) *Both quartics agree modulo 2:  $P_{15} \equiv P_{30} \equiv x^4 + x^3 + 1 \pmod{2}$  — an irreducible polynomial over  $\mathbf{F}_2$  (no roots in  $\mathbf{F}_2$ ; division by the only irreducible quadratic  $x^2 + x + 1$  leaves a nonzero remainder) — the machine fingerprint of the degree.* (c) *Isolation: the roots of  $P_{15}$  are  $1.827091\dots$ ,  $1.338261\dots$ ,  $-0.209057\dots$ ,  $-1.956295\dots$  — the largest root singles out the principal value  $2 \cos(2\pi/15)$  uniquely.*

*Proof.* (a) — direct expansion: the tower arithmetic of  $\mathbb{Z}[\sqrt{5}][\beta]$  multiplies pairs  $(A, B)$  ( $A + B\beta$  with  $A, B \in \mathbb{Z}[\sqrt{5}]$ ) by the rule  $\beta^2 = D$  with an integral element  $D = 30 \mp 6\sqrt{5}$ ; all intermediate results are integral because  $\beta^2 \in \mathbb{Z}[\sqrt{5}]$  and all coefficients are integral. The laboratory certificate (`_bch_exact_residual`) returns the zero tower element — that is the identity. (b) — reduction of coefficients:  $(1, -1, -4, 4, 1) \bmod 2 = (1, 1, 0, 0, 1)$  and  $(1, 1, -4, -4, 1) \bmod 2 = (1, 1, 0, 0, 1)$ ; the irreducibility of  $x^4 + x^3 + 1$  over  $\mathbf{F}_2$  is standard. (c) — the numeric check at dps 60: the radical forms match  $2 \cos$  with relative error 0 within precision; the ordering of the roots is strict.  $\square$

*Remark 6.* The quartics  $P_{15}, P_{30}$  are norms of the cyclic quartic field of discriminant 1125; their reflected pair  $P_{30}(x) = P_{15}(-x)$  is the arithmetic echo of the fact that the levels 15 and 30 share one cyclotomic field  $\mathbb{Q}(\zeta_{15}) = \mathbb{Q}(\zeta_{30})$  (monograph 16/20, theorem H1). The constructibility of the level (Gauss–Wantzel) and

the cyclicity  $C_4$  are two faces of the same coin: the tower  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{5}) \subset K$  has exactly one intermediate floor.

**Protocol data:** The exact layer v1.2 — the numbers

Radicals:  $b_{\text{Ch}}(15) = \frac{7-\sqrt{5}-\sqrt{30-6\sqrt{5}}}{8} = 0.086454542357399104497872428014682822\dots$ ;  
 $b_{\text{Ch}}(30) = \frac{9-\sqrt{5}-\sqrt{30+6\sqrt{5}}}{8} = 0.021852399266194362071433252130400468\dots$  (the match with  $1 - \cos$  at dps 60 is 0 within precision). Minimal quartics:  $x^4 - x^3 - 4x^2 + 4x + 1$  and  $x^4 + x^3 - 4x^2 - 4x + 1$ ; the reflection  $P_{30}(x) = P_{15}(-x)$ ; the tower residual = 0 (an integer identity); GF(2): both reduce to  $x^4 + x^3 + 1$  (irreducible); the pairing  $\beta\beta' = 12\sqrt{5}$ ;  $\beta^2 + \beta'^2 = 60$ ; the field  $K = \mathbb{Q}(\zeta_{15})^+$ ,  $\text{Gal} \cong C_4$ ,  $\text{disc } K = 1125 = 3^2 \cdot 5^3 = 5 \cdot 15 \cdot 15$  — the same number 1125 as the volume  $\text{vol}_h$  of monograph 16/20.

**Summary of results:** summary

The exact layer of v1.2 is raised to the level of theorems: the closed radicals of both constructible rungs, the minimal quartics derived from first principles (Ramanujan sums + the tower norm + Newton + the reflection), the Galois structure  $C_4$  with the unique quadratic floor  $\mathbb{Q}(\sqrt{5})$ , the integer tower certificate and the GF(2) fingerprint. One field — two rungs.

**Verification:** how to reproduce

python3 laboratory.py → “Designer → item 5” (radicals + exact/numeric verdicts,  $N = 15/30$ ); batch mode: a scenario with the bch items ( $N = 15, N = 30$ ); tests: `pytest tests/ -k bch`; Lean: the `bch_radicals` block; `--check-baseline`.